

Reviewer Pack — Minimal Topological Entropy and Resolution Proof Width

Entry 1f0c74c7c7a2 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Topological Entropy and Resolution Proof Width

Entry ID: 1f0c74c7c7a2 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-07 10:31:13 UTC

1. Conjecture

Field A (mathematical branch): Topological Dynamics **Field B** (complexity object): Boolean Satisfiability (Resolution Proof Complexity)

Statement:

The minimal topological entropy of the orbit space induced by the satisfiability problem of a CNF formula φ is bounded by its resolution proof width, i.e., $h(\text{OrbitSpace}(\varphi)) = \Theta(w(\varphi))$.

Rationale (proposer's reasoning):

Topological dynamics provides a rich framework for studying the behavior of systems. If this conjecture holds, it suggests that the complexity of satisfiability problems can be understood through the lens of dynamical systems, potentially revealing new insights into the structure of resolution proof trees.

Taxonomy category: Topological_Dynamics (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): d451d6e9fe2dc8f2

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if at least 80% of the seeds yield a correlation coefficient (r) ≥ 0.7 and the mean minimal topological entropy $h(\text{OrbitSpace}(\varphi))$ is within a factor of 2 from the mean resolution proof width $w(\varphi)$. The conjecture is falsified if any seed results in an $r < 0.5$ or $h(\text{OrbitSpace}(\varphi)) / w(\varphi) > 2$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 8 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal Topological Entropy AND Boolean Satisfiability Resolution Proof Width - CNF Formula Orbit Space Entropy Bounded by Resolution Proof Width - Entropy of Satisfiability Problems in Topological Dynamics related to Resolution Complexity

Top relevant hits considered: - [http://arxiv.org/abs/1807.02622v2] Rényi Entropy Power Inequalities via Normal Transport and Rotation - [http://arxiv.org/abs/2112.13763v3] Yet Another Proof of the Joint Convexity of Relative Entropy - [http://arxiv.org/abs/2403.07054v3] Minimal Fractional Topological Insulator in half-filled conjugate moiré Chern bands - [http://arxiv.org/abs/0804.4433v1] SPACE: the SPECTROSCOPIC All-sky Cosmic Explorer - [http://arxiv.org/abs/2005.12856v2] Topological Entropy for Arbitrary Subsets of Infinite Product Spaces - [http://arxiv.org/abs/1905.10867v2] Regular resolution for CNFs with almost bounded one-sided treewidth - [http://arxiv.org/abs/2110.06573v2] A Parametric and Feasibility Study for Data Sampling of the Dynamic Mode Decomposition-Range, Resolution, and Universal - [http://arxiv.org/abs/1804.02478v5] Complexity of the CNF-satisfiability problem

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.randint(-n, -1), random.randint(1, n)]
            if random.choice([True, False]):
                clause = [-x for x in clause]
            clauses.append(clause)
        return clauses

    def topological_sort(clauses):
        graph = {i: [] for i in range(1, 2 * n + 1)}
        in_degree = {i: 0 for i in range(1, 2 * n + 1)}

        for clause in clauses:
            for literal in clause:
                if literal > 0:
                    graph[literal].append(-literal)
```

```

        in_degree[-literal] += 1
    else:
        graph[-literal].append(literal)
        in_degree[literal] += 1

queue = [i for i, degree in in_degree.items() if degree == 0]
topo_order = []

while queue:
    node = queue.pop(0)
    topo_order.append(node)

    for neighbor in graph[node]:
        in_degree[neighbor] -= 1
        if in_degree[neighbor] == 0:
            queue.append(neighbor)

return topo_order

def compute_orbit_space(clauses):
    topo_order = topological_sort(clauses)
    orbit_space = set()
    for literal in topo_order:
        orbit_space.add(literal)
    return len(orbit_space)

def resolution_width(clauses):
    n = len(clauses) // 2
    clauses = [tuple(sorted(c)) for c in clauses]
    clauses = list(set(clauses))

    def resolve(clause1, clause2):
        new_clauses = []
        for literal in clause1:
            if -literal in clause2:
                continue
            new_clause = set(clause1)
            new_clause.remove(literal)
            new_clause.update(clause2)
            new_clauses.append(tuple(sorted(new_clause)))
        return new_clauses

    queue = clauses[:]
    while queue:
        new_queue = []
        for i, clause1 in enumerate(queue):
            for j, clause2 in enumerate(queue):
                if i >= j:
                    continue
                new_clauses = resolve(clause1, clause2)
                for new_clause in new_clauses:
                    if new_clause not in queue and new_clause not in new_queue:
                        new_queue.append(new_clause)
        queue.extend(new_queue)

    return len(queue)

```

```

n = 30
cnf_formula = generate_cnf(n)
orbit_space_size = compute_orbit_space(cnf_formula)
resolution_width_value = resolution_width(cnf_formula)

if orbit_space_size == 0 or resolution_width_value == 0:
    return {
        "metric_name": "orbit_space_resolution_ratio",
        "metric_value": None,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": "empty_orbit_space"
    }

metric_value = orbit_space_size / resolution_width_value
return {
    "metric_name": "orbit_space_resolution_ratio",
    "metric_value": metric_value,
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": True,
    "counterexample": ""
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, \"metric_name\": \"{trial_result['metric_name']}\", \"n_max\": {n_max}, \"conjecture_holds\": {conjecture_holds}, \"counterexample\": \"{counterexample}\"}}")
        results.append(trial_result)

    if all(result["metric_value"] is not None for result in results):
        mean_metric = sum(result["metric_value"] for result in results) / len(results)
        std_metric = math.sqrt(sum((result["metric_value"] - mean_metric) ** 2 for result in results) / len(results))
        support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

        if support_fraction >= 0.8:
            print(f"RESULT: SUPPORTED mean={mean_metric} std={std_metric} support_fraction={support_fraction}")
        else:
            first_failing_seed = next(seed for seed, result in enumerate(results) if not result["conjecture_holds"])
            print(f"RESULT: FALSIFIED counterexample=\"orbit_space_resolution_ratio\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE missing_data")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

```

Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_23bf92cd.py", line 125, in <module>
    trial_result = run_trial(seed)
                    ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_23bf92cd.py", line 97, in run_trial
    orbit_space_size = compute_orbit_space(cnf_formula)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_23bf92cd.py", line 58, in compute_orbit
    topo_order = topological_sort(clauses)
                  ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_23bf92cd.py", line 41, in topological_s
    in_degree[literal] += 1
    ~~~~~^~~~~~
KeyError: -16

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means it did not complete its execution to provide a result for the conjecture. | next: Investigate and fix the crash in the test code to ensure it can run to completion and provide results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	20107
2	preregistration	ollama_remote	glm4:latest	0	0	13140
3	novelty	ollama_remote	glm4:latest	0	0	8621
4	novelty	ollama_remote	glm4:latest	0	0	10079
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18850
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10767
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11719
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13303
9	judge	ollama_remote	glm4:latest	0	0	15717

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 122302 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/1f0c74c7c7a2.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/1f0c74c7c7a2.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/1f0c74c7c7a2.tar.gz` (if generated)